Informatyka niestacjonarnie Alex Jankowiak Raport systemy operacyjne – laboratoria 16.05.2026

Temat 5. Linux/Unix – zaawansowane wyszukiwanie plików

Ad. Zadanie 12

Rezultat komendy find /usr –name ‘*.txt’

```
/usr/lib/python3/dist-packages/python_debian-1.0.1.dist-info/top_level.txt
/usr/lib/python3/dist-packages/python_debianbts-4.1.1.dist-info/entry_points.txt
/usr/lib/python3/dist-packages/python_debianbts-4.1.1.dist-info/top_level.txt
/usr/lib/python3/dist-packages/apt_listchanges-4.8.dist-info/entry_points.txt
/usr/lib/python3/dist-packages/apt_listchanges-4.8.dist-info/top_level.txt
/usr/lib/python3/dist-packages/chardet-5.2.0.dist-info/entry_points.txt
/usr/lib/python3/dist-packages/chardet-5.2.0.dist-info/top_level.txt
/usr/lib/python3/dist-packages/charset_normalizer-3.4.2.dist-info/entry_points.txt
/usr/lib/python3/dist-packages/charset_normalizer-3.4.2.dist-info/top_level.txt
/usr/lib/python3/dist-packages/certifi-2025.1.31.egg-info/top_level.txt
/usr/lib/python3/dist-packages/certifi-2025.1.31.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/reportbug-13.2.0.egg-info/top_level.txt
/usr/lib/python3/dist-packages/reportbug-13.2.0.egg-info/dependency_links.txt
/usr/lib/python3/dist-packages/requests-2.32.3.dist-info/top_level.txt
/usr/lib/python3/dist-packages/python_apt-3.0.0.egg-info/top_level.txt
/usr/lib/python3/dist-packages/python_apt-3.0.0.egg-info/dependency_links.txt
/usr/lib/python3.13/LICENSE.txt
find: '/usr/lost+found': Permission denied
/usr/share/doc/reportbug/HowToReportGoodBugs.txt
/usr/share/doc/debian/bug-mailserver-refcard.txt
/usr/share/doc/debian/source-unpack.txt
/usr/share/doc/debian/bug-log-access.txt
/usr/share/doc/mount/mount.txt
/usr/share/doc/busybox/syslog.conf.txt
/usr/share/doc/util-linux/00-about-docs.txt
/usr/share/doc/util-linux/pg.txt
/usr/share/doc/util-linux/getopt_changelog.txt
/usr/share/doc/util-linux/getopt.txt
/usr/share/doc/util-linux/hwclock.txt
/usr/share/doc/util-linux/mount.txt
/usr/share/doc/util-linux/deprecated.txt
/usr/share/doc/util-linux/howto-compilation.txt
/usr/share/doc/util-linux/col.txt
/usr/share/doc/util-linux/howto-build-sys.txt
/usr/share/doc/util-linux/blkid.txt
/usr/share/doc/util-linux/howto-man-page.txt
/usr/share/doc/util-linux/cal.txt
/usr/share/doc/util-linux/howto-tests.txt
/usr/share/doc/util-linux/modems-with-agetty.txt
/usr/share/doc/util-linux/howto-debug.txt
/usr/share/doc/util-linux/release-schedule.txt
/usr/share/doc/util-linux/PAM-configuration.txt
/usr/share/doc/python3-debianbts/THANKS.txt
/usr/share/doc/publicsuffix/examples/test_psl.txt
/usr/share/doc/libdb5.3t64/build_signature_arm64.txt
/usr/share/doc/openssl/fingerprints.txt
/usr/share/vim/vim91/doc/uganda.txt
/usr/share/vim/vim91/doc/version9.txt
/usr/share/vim/vim91/doc/help.txt
/usr/share/vim/vim91/doc/sponsor.txt
/usr/share/perl/5.40.1/Unicode/Collate/keys.txt
/usr/share/perl/5.40.1/Unicode/Collate/allkeys.txt
/usr/share/perl/5.40.1/unicore/Blocks.txt
/usr/share/perl/5.40.1/unicore/SpecialCasing.txt
/usr/share/perl/5.40.1/unicore/NamedSequences.txt
find: '/usr/local/lost+found': Permission denied
student@pantera-ajankowiak:~$ _
```

Ad. Zadanie 13

```
/usr/share/nano/yaml.nanorc
/usr/share/nano/lua.nanorc
/usr/share/nano/nanohelp.nanorc
/usr/share/nano/ruby.nanorc
/usr/share/nano/guile.nanorc
/usr/share/nano/asm.nanorc
/usr/share/nano/rust.nanorc
/usr/share/nano/texinfo.nanorc
/usr/share/nano/man.nanorc
/usr/share/nano/awk.nanorc
/usr/share/nano/java.nanorc
/usr/share/nano/elisp.nanorc
/usr/share/nano/tex.nanorc
/usr/share/nano/xml.nanorc
/usr/share/nano/perl.nanorc
/usr/share/nano/cmake.nanorc
/usr/share/nano/debian.nanorc
/usr/share/nano/autoconf.nanorc
/usr/share/nano/json.nanorc
/usr/share/nano/javascript.nanorc
/usr/share/openssh
/usr/share/openssh/sshd_config.md5sum
/usr/share/openssh/sshd_config
/usr/share/grub
/usr/share/grub/euro.pf2
/usr/share/grub/ascii.h
/usr/share/grub/widthspec.h
/usr/share/grub/ascii.pf2
/usr/share/grub/unicode.pf2
/usr/share/grub/grub-mkconfig_lib
/usr/share/grub/default
/usr/share/grub/default/grub.md5sum
/usr/share/grub/default/grub
/usr/local
/usr/local/sbin
/usr/local/bin
/usr/local/share
/usr/local/share/ca-certificates
/usr/local/share/man
/usr/local/lost+found
find: '/usr/local/lost+found': Permission d
/usr/local/libexec
/usr/local/include
/usr/local/etc
/usr/local/src
/usr/local/man
/usr/local/games
/usr/local/lib
/usr/local/lib/python3.13
/usr/local/lib/python3.13/dist-packages
/usr/games
/usr/src
/usr/include
/usr/include/iproute2
/usr/include/iproute2/bpf_elf.h
/usr/include/clif.h
student@pantera-ajankowiak:~$ _
```

Ad. Zadanie 14

```
/usr/share/man/pl/man4
/usr/share/man/pl/man1
/usr/share/man/pl/man7
/usr/share/man/pl/man6
/usr/share/man/pl/man2
/usr/share/man/pl/man3
/usr/share/man/cs
/usr/share/man/cs/man5
/usr/share/man/cs/man8
/usr/share/man/cs/man1
/usr/share/man/ja
/usr/share/man/ja/man5
/usr/share/man/ja/man8
/usr/share/man/ja/man1
/usr/share/man/ru
/usr/share/man/ru/man5
/usr/share/man/ru/man8
/usr/share/man/ru/man1
/usr/share/man/tr
/usr/share/man/tr/man5
/usr/share/man/tr/man8
/usr/share/man/tr/man1
/usr/share/man/sv
/usr/share/man/sv/man5
/usr/share/man/sv/man8
/usr/share/man/sv/man1
/usr/share/man/sv/man3
/usr/share/man/nl
/usr/share/man/nl/man5
/usr/share/man/nl/man8
/usr/share/man/nl/man1
/usr/share/man/nl/man7
/usr/share/man/fi
/usr/share/man/fi/man1
/usr/share/man/id
/usr/share/man/id/man5
/usr/share/man/id/man8
/usr/share/man/id/man1
/usr/share/man/sl
/usr/share/man/sl/man8
/usr/share/man/sl/man1
/usr/share/man/man7
/usr/share/man/pt_BR
/usr/share/man/pt_BR/man5
/usr/share/man/pt_BR/man8
/usr/share/man/pt_BR/man1
/usr/share/man/hu
/usr/share/man/hu/man5
/usr/share/man/hu/man1
/usr/share/man/man6
/usr/share/man/zh_CN
/usr/share/man/zh_CN/man5
/usr/share/man/zh_CN/man8
/usr/share/man/zh_CN/man1
/usr/share/man/man2
/usr/share/man/man3
student@pantera-ajankowiak:~$
```

Ad. Zadanie 15

Opcje dla type to: b, c, d, l, p, f, s

Ad. Zadanie 16

```
student@pantera-ajankowiak:~$ find /usr -size 0
/usr/lib/python3/dist-packages/charset_normalizer/py.typed
/usr/lib/python3/dist-packages/debian/py.typed
/usr/lib/python3/dist-packages/apt/py.typed
/usr/lib/python3/dist-packages/apt_listchanges/__init__.py
/usr/lib/python3/dist-packages/urllib3/contrib/__init__.py
/usr/lib/python3/dist-packages/debianbts/py.typed
/usr/lib/python3/dist-packages/certifi/py.typed
/usr/lib/python3/dist-packages/idna/py.typed
/usr/lib/python3/dist-packages/chardet/metadata/__init__.py
/usr/lib/python3/dist-packages/chardet/cli/__init__.py
/usr/lib/python3/dist-packages/chardet/py.typed
/usr/lib/python3.13/email/mime/__init__.py
/usr/lib/python3.13/pydoc_data/__init__.py
/usr/lib/python3.13/test/libregrtest/__init__.py
/usr/lib/python3.13/test/typinganndata/__init__.py
/usr/lib/python3.13/urllib/__init__.py
/usr/lib/grub/arm64-efi/fdt.lst
find: '/usr/lost+found': Permission denied
/usr/share/X11/locale/isiri-3342/Compose
/usr/share/X11/locale/iso8859-11/Compose
/usr/share/X11/locale/cs_CZ.UTF-8/XLC_LOCALE
/usr/share/X11/locale/cs_CZ.UTF-8/XI18N_OBJS
/usr/share/X11/locale/el_GR.UTF-8/XLC_LOCALE
/usr/share/X11/locale/el_GR.UTF-8/XI18N_OBJS
/usr/share/X11/locale/nokhchi-1/Compose
/usr/share/X11/locale/tscii-0/Compose
/usr/share/X11/locale/microsoft-cp1256/Compose
/usr/share/X11/locale/microsoft-cp1255/Compose
/usr/share/X11/locale/th_TH/Compose
/usr/share/X11/locale/am_ET.UTF-8/XLC_LOCALE
/usr/share/X11/locale/am_ET.UTF-8/XI18N_OBJS
/usr/share/X11/locale/km_KH.UTF-8/XLC_LOCALE
/usr/share/X11/locale/sr_RS.UTF-8/XLC_LOCALE
/usr/share/X11/locale/tatar-cyr/Compose
/usr/share/X11/locale/microsoft-cp1251/Compose
/usr/share/X11/locale/fi_FI.UTF-8/XLC_LOCALE
/usr/share/X11/locale/fi_FI.UTF-8/XI18N_OBJS
/usr/share/X11/locale/C/Compose
/usr/share/X11/locale/iscii-dev/Compose
/usr/share/runit/meta/ssh/installed
/usr/share/xml/iso-codes/iso_3166-3.xml
/usr/share/dictionaries-common/site-elisp/.nosearch
/usr/share/dictionaries-common/elanguages
find: '/usr/local/lost+found': Permission denied
student@pantera-ajankowiak:~$
```

Ad. Zadanie 17

```
/usr/share/X11/xkb/rules/evdev.lst
/usr/share/X11/xkb/rules/base
/usr/share/X11/xkb/rules/evdev.extras.xml
/usr/share/X11/xkb/rules/base.lst
/usr/share/X11/xkb/rules/evdev
/usr/share/X11/xkb/rules/base.extras.xml
/usr/share/X11/xkb/geometry/pc
/usr/share/X11/xkb/geometry/macintosh
/usr/share/X11/XErrorDB
/usr/share/i18n/locales/km_KH
/usr/share/i18n/locales/translit_compat
/usr/share/i18n/locales/uk_UA
/usr/share/i18n/locales/am_ET
/usr/share/i18n/locales/translit_combining
/usr/share/i18n/locales/gez_ER@abegede
/usr/share/i18n/locales/translit_font
/usr/share/i18n/locales/translit_cjk_compat
/usr/share/i18n/locales/ko_KR
/usr/share/i18n/locales/lo_LA
/usr/share/i18n/charmaps/SHIFT_JIS.gz
/usr/share/i18n/charmaps/WINDOWS-31J.gz
/usr/share/i18n/charmaps/EUC-JISX0213.gz
/usr/share/i18n/charmaps/BIG5.gz
/usr/share/i18n/charmaps/EUC-KR.gz
/usr/share/i18n/charmaps/GB2312.gz
/usr/share/i18n/charmaps/EUC-JP-MS.gz
/usr/share/i18n/charmaps/EUC-JP.gz
/usr/share/i18n/charmaps/SHIFT_JISX0213.gz
/usr/share/iso-codes/json/iso_3166-1.json
/usr/share/iso-codes/json/iso_639-2.json
/usr/share/common-licenses/LGPL-2.1
/usr/share/common-licenses/MPL-1.1
/usr/share/common-licenses/GPL-3
/usr/share/groff/1.23.0/tmac/hyphen.sv
/usr/share/groff/1.23.0/tmac/hyphen.en
/usr/share/groff/1.23.0/tmac/mdoc/doc-common
/usr/share/groff/1.23.0/tmac/mdoc/doc-syms
/usr/share/groff/1.23.0/tmac/an.tmac
/usr/share/groff/1.23.0/tmac/www.tmac
/usr/share/groff/1.23.0/tmac/rfc1345.tmac
/usr/share/groff/1.23.0/tmac/hyphen.fr
/usr/share/groff/1.23.0/tmac/doc-old.tmac
/usr/share/keyrings/debian-archive-keyring.pgp
/usr/share/keyrings/debian-archive-removed-keys.pgp
/usr/share/perl5/Debian/DictionariesCommon.pm
/usr/share/console-setup/KeyboardNames.pl
/usr/share/xml/iso-codes/iso_3166-1.xml
/usr/share/xml/iso-codes/iso_4217.xml
/usr/share/xml/iso-codes/iso_639-2.xml
/usr/share/publicsuffix/public_suffix_list.dafsa
/usr/share/dictionaries-common/site-elisp/flyspell.el
/usr/share/initramfs-tools/hook-functions
/usr/share/gcc/python/libstdcxx/v6/xmethods.py
/usr/share/grub/euro.pf2
/usr/share/grub/widthspec.h
find: '/usr/local/lost+found': Permission denied
student@pantera-ajankowiak:~$
```

Ad. Zadanie 18

c - bajty

w - 2-bajtowe słowa

k - KiB (1024 bajty)

M - MiB (1024 * KiB)

G - GiB (1024 * MiB)

Ad. Zadanie 19

```
student@pantera-ajankowiak:~$ find /usr -size +10M
/usr/lib/udev/hwdb.bin
find: '/usr/lost+found': Permission denied
/usr/share/dict/polish
find: '/usr/local/lost+found': Permission denied
student@pantera-ajankowiak:~$
```

są to /usr/lib/udev/hwdb.bin i /usr/share/dict/polish

Ad. Zadanie 20

```
student@pantera-ajankowiak:~$ find /usr -size +10M -exec ls -l {} \;
-r--r--r-- 1 root root 13200416 Apr 12 16:00 /usr/lib/udev/hwdb.bin
find: '/usr/lost+found': Permission denied
-rw-r--r-- 1 root root 61821278 Sep  2  2024 /usr/share/dict/polish
find: '/usr/local/lost+found': Permission denied
student@pantera-ajankowiak:~$
```

Ad. Zadanie 21

```
student@pantera-ajankowiak:~$ mkdir ~/KOPIA
student@pantera-ajankowiak:~$ find /usr -size +1M -size -5M -exec cp --parents {} ~/KOPIA\;
find: missing argument to `-exec'
student@pantera-ajankowiak:~$ find /usr -size +1M -size -5M -exec cp --parents {} ~/KOPIA \;
find: '/usr/lost+found': Permission denied
find: '/usr/local/lost+found': Permission denied
student@pantera-ajankowiak:~$ _
```

Ad. Zadanie 22

find /home -mtime -7

```
/home/student/KOPIA/usr/share/perl/5.40.1/unicore/Name.pl
/home/student/KOPIA/usr/share/perl/5.40.1/Module/CoreList.pm
/home/student/KOPIA/usr/share/perl/5.40.1/Unicode/Collate/allkeys.txt
/home/student/KOPIA/usr/share/doc/linux-image-6.12.74+deb13+1-arm64/changelog.gz
/home/student/KOPIA/usr/share/doc/linux-image-6.12.73+deb13-arm64/changelog.gz
/home/student/KOPIA/usr/share/ispell/polish.mwl.gz
/home/student/KOPIA/usr/share/vim/vim91/doc/version9.txt
/home/student/KOPIA/usr/share/i18n/locales/iso14651_t1_pinyin
/home/student/KOPIA/usr/share/i18n/locales/iso14651_t1_common
/home/student/KOPIA/usr/share/grub/unicode.pf2
/home/student/KOPIA/usr/share/misc/pci.ids
/home/student/KOPIA/usr/sbin/grub-install
/home/student/KOPIA/usr/sbin/apparmor_parser
/home/student/KOPIA/usr/lib/grub/arm64-efi/monolithic/grubaa64.efi
/home/student/KOPIA/usr/lib/grub/arm64-efi/monolithic/gcdaa64.efi
/home/student/KOPIA/usr/lib/grub/arm64-efi/monolithic/grubnetaa64.efi
/home/student/KOPIA/usr/lib/grub/arm64-efi/monolithic/grubnetaa64-installer.efi
/home/student/KOPIA/usr/lib/grub/arm64-efi-signed/grubaa64.efi.signed
/home/student/KOPIA/usr/lib/grub/arm64-efi-signed/gcdaa64.efi.signed
/home/student/KOPIA/usr/lib/grub/arm64-efi-signed/grubnetaa64.efi.signed
/home/student/KOPIA/usr/lib/grub/arm64-efi-signed/grubnetaa64-installer.efi.signed
/home/student/KOPIA/usr/lib/systemd/systemd-networkd
/home/student/KOPIA/usr/lib/udev/hwdb.d/20-pci-vendor-model.hwdb
/home/student/KOPIA/usr/lib/udev/hwdb.d/20-OUI.hwdb
/home/student/KOPIA/usr/lib/udev/hwdb.d/20-usb-vendor-model.hwdb
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/kernel/drivers/gpu/drm/amd/amdgpu/amdgpu.ko.xz
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/modules.alias
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/modules.alias.bin
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers/gpu/drm/amd/amdgpu/amdgpu.ko.xz
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/modules.alias
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/modules.alias.bin
/home/student/KOPIA/usr/lib/locale/locale-archive
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Unicode/Collate/Collate.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/CN/CN.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/JP/JP.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/TW/TW.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/KR/KR.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libc.so.6
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libsqlite3.so.0.8.6
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libssl.so.3
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libdns-9.20.21-1~deb13u1-Debian.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libapt-pkg.so.7.0.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libnftables.so.1.1.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libgnutls.so.30.40.3
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libslang.so.2.3.3
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.33
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libperl.so.5.40.1
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/systemd/libsystemd-core-257.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libunistring.so.5.2.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libxml2.so.2.9.14
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libp11-kit.so.0.4.1
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libX11.so.6.4.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libdb-5.3.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libsystemd.so.0.40.0
/home/student/KOPIA/usr/lib/openssh/sshd-session
/home/student/KOPIA/usr/lib/openssh/sshd-auth
student@pantera-ajankowiak:~$
```

```
student@pantera-ajankowiak:~$ find /home -atime +14
/home/lost+found
find: '/home/lost+found': Permission denied
/home/student/.lesshst
/home/student/.bash_logout
find: '/home/student/A/B': Permission denied
find: '/home/student/A/a.txt': Permission denied
student@pantera-ajankowiak:~$ _
```

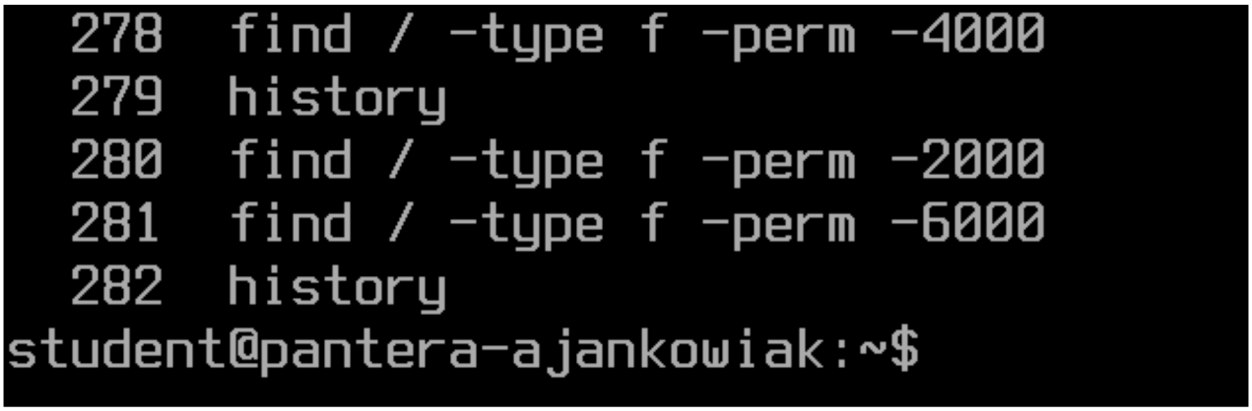
find /home -ctime 0:

```
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/kernel/drivers/gpu
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/kernel/drivers/gpu/drm
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/kernel/drivers/gpu/drm/amd
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/kernel/drivers/gpu/drm/amd/
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/kernel/drivers/gpu/drm/amd/
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/modules.alias
/home/student/KOPIA/usr/lib/modules/6.12.74+deb13+1-arm64/modules.alias.bin
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers/gpu
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers/gpu/drm
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers/gpu/drm/amd
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers/gpu/drm/amd/am
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/kernel/drivers/gpu/drm/amd/am
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/modules.alias
/home/student/KOPIA/usr/lib/modules/6.12.73+deb13-arm64/modules.alias.bin
/home/student/KOPIA/usr/lib/locale
/home/student/KOPIA/usr/lib/locale/locale-archive
/home/student/KOPIA/usr/lib/aarch64-linux-gnu
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Unicode
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Unicode/Collate
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Unicode/Collate/Collat
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/CN
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/CN/CN.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/JP
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/JP/JP.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/TW
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/TW/TW.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/KR
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/perl/5.40.1/auto/Encode/KR/KR.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libc.so.6
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libsqlite3.so.0.8.6
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libssl.so.3
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libdns-9.20.21-1~deb13u1-Debian.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libapt-pkg.so.7.0.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libnftables.so.1.1.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libgnutls.so.30.40.3
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libslang.so.2.3.3
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libstdc++.so.6.0.33
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libperl.so.5.40.1
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/systemd
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/systemd/libsystemd-core-257.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libunistring.so.5.2.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libxml2.so.2.9.14
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libp11-kit.so.0.4.1
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libX11.so.6.4.0
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libdb-5.3.so
/home/student/KOPIA/usr/lib/aarch64-linux-gnu/libsystemd.so.0.40.0
/home/student/KOPIA/usr/lib/openssh
/home/student/KOPIA/usr/lib/openssh/sshd-session
/home/student/KOPIA/usr/lib/openssh/sshd-auth
student@pantera-ajankowiak:~$ _
```

Ad. Zadanie 23

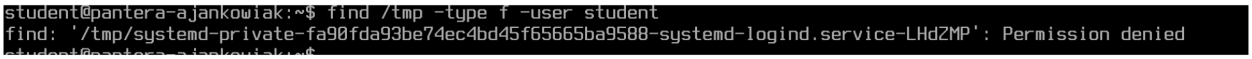
- -mmin - modyfikacja zawartości pliku podana w minutach (odpowiednik -mtime, ale w minutach).
- -amin - ostatni dostęp do pliku w minutach (odpowiednik -atime).
- -cmin - zmiana statusu (metadanych) pliku w minutach (odpowiednik -ctime).

Ad. Zadanie 24



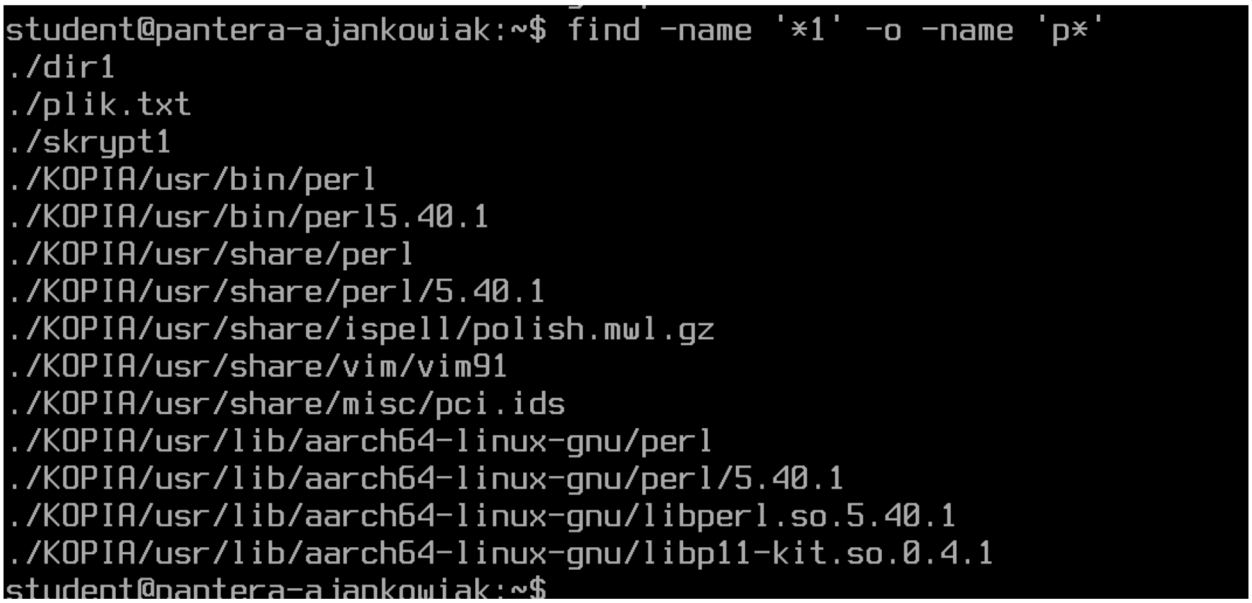
wyświetliło się bardzo dużo plików z brakiem dostępu

Ad. Zadanie 25



Ad. Zadanie 26

pliki których nazwy kończą się na 1 lub zaczynają na p



pliki których nazwy kończą się na 1 i równocześnie zaczynają na p (można jako 'p*1')

```
student@pantera-ajankowiak:~$ find -name '*1' -a -name 'p*'
./KOPIA/usr/bin/perl5.40.1
student@pantera-ajankowiak:~$ _
```

Ad. Zadanie 27

```
-rwxrwxrwx 1 student student    0 May 16 13:08 a.txt
student@pantera-ajankowiak:~$ find /home/student -perm -g=x -exec chmod o+x {} +
student@pantera-ajankowiak:~$ ls -l /home/student
total 20
drwxrwxrwx 3 student student 4096 May 16 13:08 A
drwxrwxrwx 3 student student 4096 May 16 14:08 KOPIA
drwxrwxrwx 2 student student 4096 May 16 13:17 dir1
-rwxrwxrwx 1 student student   12 May 16 12:55 plik.txt
-rwxrwxrwx 1 student student    0 May 16 12:30 skrypt
-rwxrwxrwx 1 student student    0 May 16 12:42 skrypt1
drwxrwxrwx 2 student student 4096 May 16 13:07 test
-rwxrwxrwx 1 student student    0 May 16 12:00 test.txt
-rwxrwxrwx 1 student audio      0 May 16 12:37 test1.txt
-rwxrwxrwx 1 student student    0 May 16 13:17 z.txt
-rwxrwxrwx 1 student student    0 May 16 13:17 z1.txt
student@pantera-ajankowiak:~$ _
```